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Mid Semester Examination March 2025

Name of the Program: B.Tech

Semester: II

Name of the Course: Basic Electrical Engineering

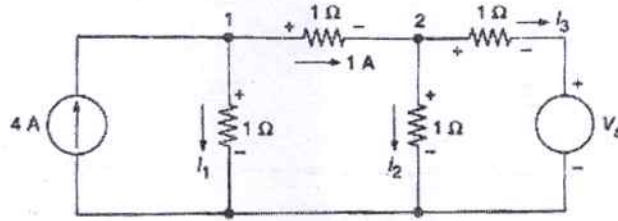
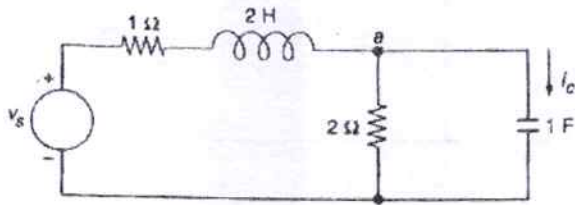
Course Code: TEE 201

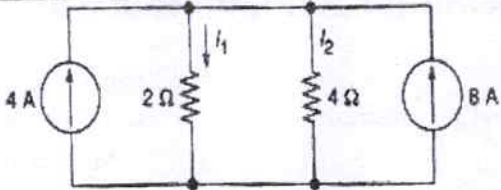
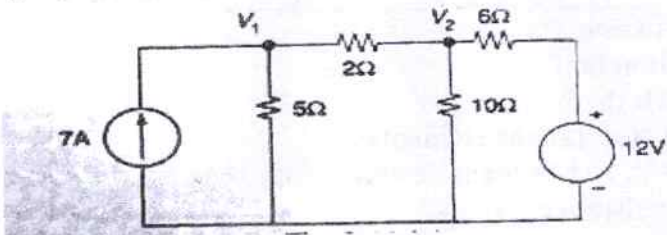
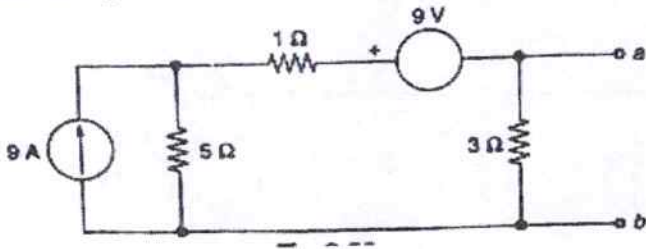
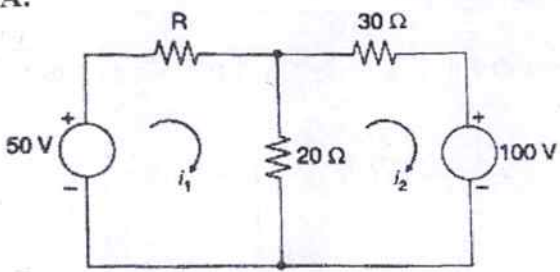
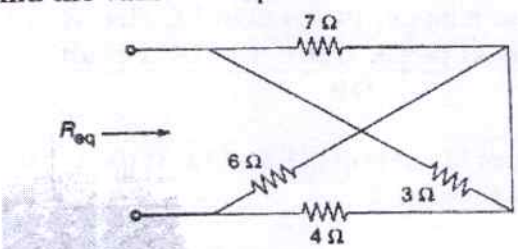
Time: 90 Minutes

Maximum Marks: 50

Note:-

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10marks)	
(a)	I. Differentiate between (i) Node and Junction (ii) Loop and Mesh (iii) Linear and Non Linear elements II. Consider a 230 V, 100 W incandescent lamp. Determine (i) The lamp resistance (ii) The lamp current (iii) The energy consumed by lamp in 8h	CO1
OR		
(b)	For the resistive circuit shown in fig. 1, using Nodal Analysis, find (a) I_1 and (b) V_s.  Fig. 1	
Q2	(10marks)	
(a)	State Thevenin Theorem. Explain the procedure to find Thevenin Equivalent along with example.	CO1, CO2
OR		
(b)	For the circuit shown in fig. 2, $i_c = -2e^{-t}$ A. Find V_s, assuming initial conditions are zero.  Fig. 2	
Q3	(10marks)	
(a)	Define maximum power transfer theorem in DC circuit. Also derive the condition of maximum power transfer in DC circuit.	CO1, CO2
OR		
(b)	For fig. 3, find the values of currents I_1 and I_2. What is the power supplied by two current sources?	

	 Fig. 3	
Q4	(10marks)	CO3
(a)	For the circuit of fig.4 (a) determine current through $2\ \Omega$ resistance using Superposition theorem.  Fig. 4(a)	
	OR	
(b)	Find the Norton equivalent of the circuit of fig 4(b) at terminals ab  Fig. 4(b)	
Q5	(10 Marsk)	CO1
(a)	For the circuit of fig. 5(a) determine the value of resistance R such that $i_1 = 0.37\text{ A}$.  Fig. 5(a)	
	OR	
(b)	In fig. 5(b), find the value of R_{eq}.  Fig. 5(b)	